



CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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Mean, Variance of Different Distributions

Distr.	Description/Example	$p_X(x), \mu, \sigma^2$
Bernoulli	x A coin toss	$p_X(x) = p^x(1-p)^{1-x}, x = 0,1$ $\mu = p, \sigma^2 = pq$
Binomial	exactly x success in n trials Prob. of 4 Heads on 10 coin flips?	$p_X(x) = \binom{n}{x} p^x q^{n-x}$ $\mu = np, \sigma^2 = npq$
Geometric	x trials required for the first success Prob. of getting first H on 10th toss?	$p_X(x) = (1-p)^{x-1} p$ $\mu = \frac{1}{p}, \sigma^2 = \frac{(1-p)}{p^2}$
Poisson	x events in a fixed time interval, when events occur independently at a constant average rate. Prob. of 3 accidents at an intersection in a day?	$p_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0,1,2, \dots$ $\mu = \lambda, \sigma^2 = \lambda$
Negative Binomial	x trials needed to obtain r successes Prob. of getting 4 th Head on 10th coin toss?	$p_X(x) = \binom{x-1}{r-1} (1-p)^{x-r} p^r$ $\mu = \frac{r}{p}, \sigma^2 = \frac{r(1-p)}{p^2}$
Hypergeometric	x success in n trials, from a population of size N , which has K successes (without replacement) Prob. of getting two Aces in 4 picks without replacement?	$p_X(x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$ $\mu = np$, where $p = K/N$ $\sigma^2 = np(1-p) \frac{N-n}{N-1}$

L12
Aug 20, 2026

Announcement

- 20th Aug: tut-3
- 21st Aug: Class tomorrow (Friday) at 10 AM
- 24th Aug: Lecture hour ~~today~~ 3 – 4 pm; tutorial 4 – 5 pm
- 25th Aug: No class on Tuesday
- ~~28th~~ Aug: Quiz-1 on ~~next Friday~~ 8 AM
31st *next-to-next Monday*

- Check course page on GitHub regularly

<https://github.com/prakashvs613/2026-CV2090-IITM/>

Higher Moments

Expectation of a function of X

$$\mu = E[X] = \sum_{x_i} x_i p_X(x_i) = \int_{-\infty}^{\infty} x f_X(x) dx$$

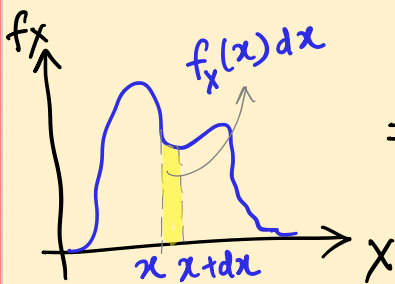
$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x) \cdot dx$$

Let $Y = g(X)$

$$E[Y] = \int_{-\infty}^{\infty} y \cdot f_Y(y) \cdot dy$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_{g(X)}(g(x)) \cdot dg(x)$$

Assume one-to-one $g(X)$



Expectation and Moments

$$\mu_{g(x)} = E[g(x)] = \int_{-\infty}^{\infty} g(x) f_x(x) dx$$

$$\mu = E[X] = \int_{-\infty}^{\infty} x f_x(x) dx$$

$$\sigma^2 = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f_x(x) dx = E[X^2] - \mu^2$$

$$m^{\text{th}} \text{ moment of } X = E[X^m]$$

$$1^{\text{st}} \text{ central moment of } X = E[(X-\mu)]$$

$$2^{\text{nd}} \text{ central moment of } X = E[(X-\mu)^2] = \sigma^2$$

$$\int_{-\infty}^{\infty} x^2 f_x(x) dx \rightarrow 2^{\text{nd}} \text{ moment of } X$$

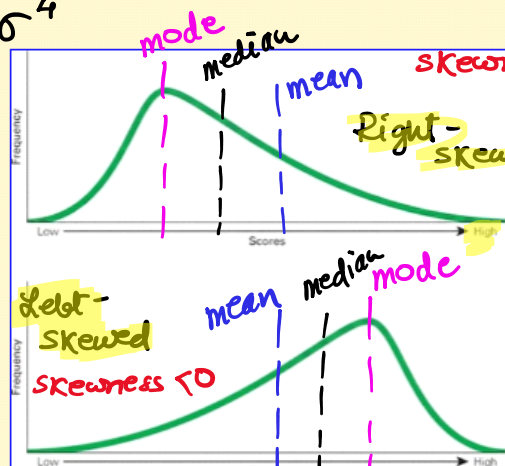
Central Moments, Skewness

$$1^{\text{st}} \text{ central moment } E[X-\mu] = 0$$

$$2^{\text{nd}} \text{ central moment } E[(X-\mu)^2] = \sigma^2$$

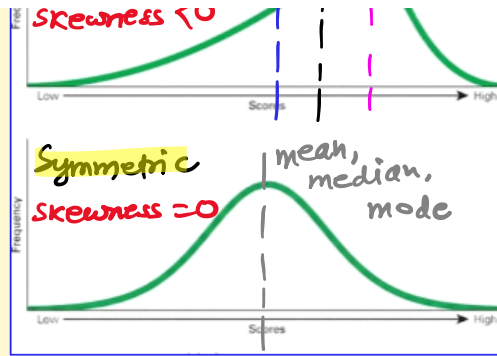
$$3^{\text{rd}} \text{ central moment } E[(X-\mu)^3] = \text{skewness}$$

$$4^{\text{th}} \text{ central moment } \frac{E[(X-\mu)^4]}{\sigma^4} = \text{kurtosis}$$



Qualitatively mark

- Mean,
 - Median, and
 - Mode,
- on all three figures.



on all three figures.

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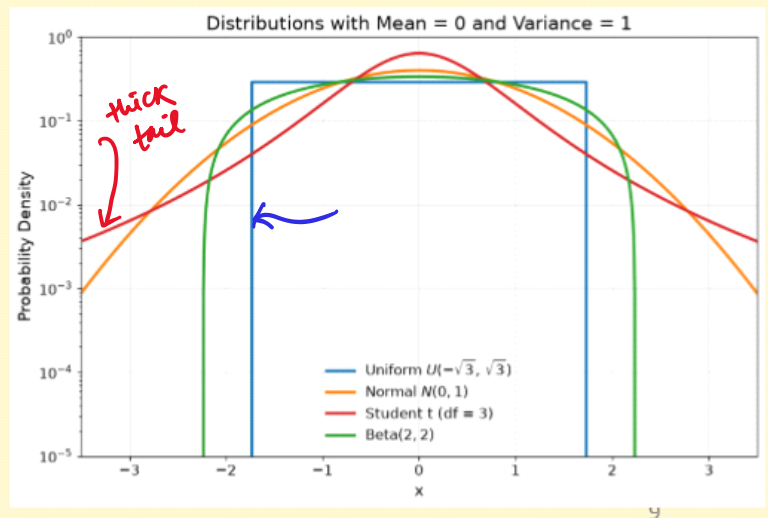
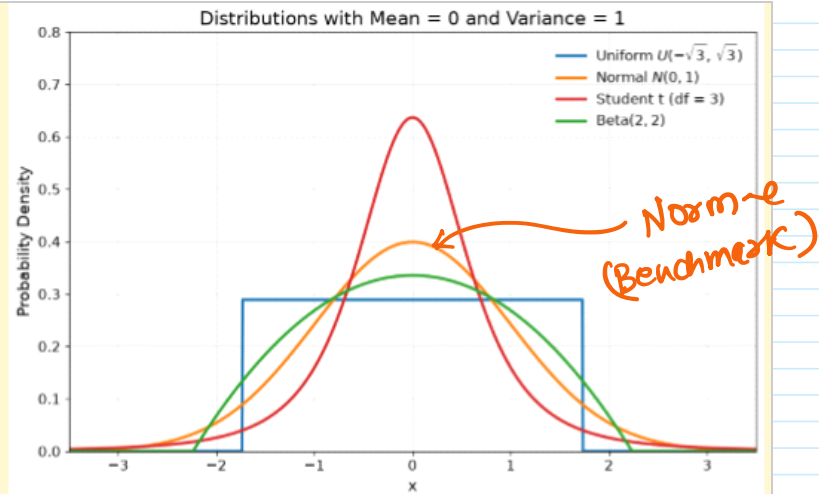
Kurtosis

Excess kurtosis

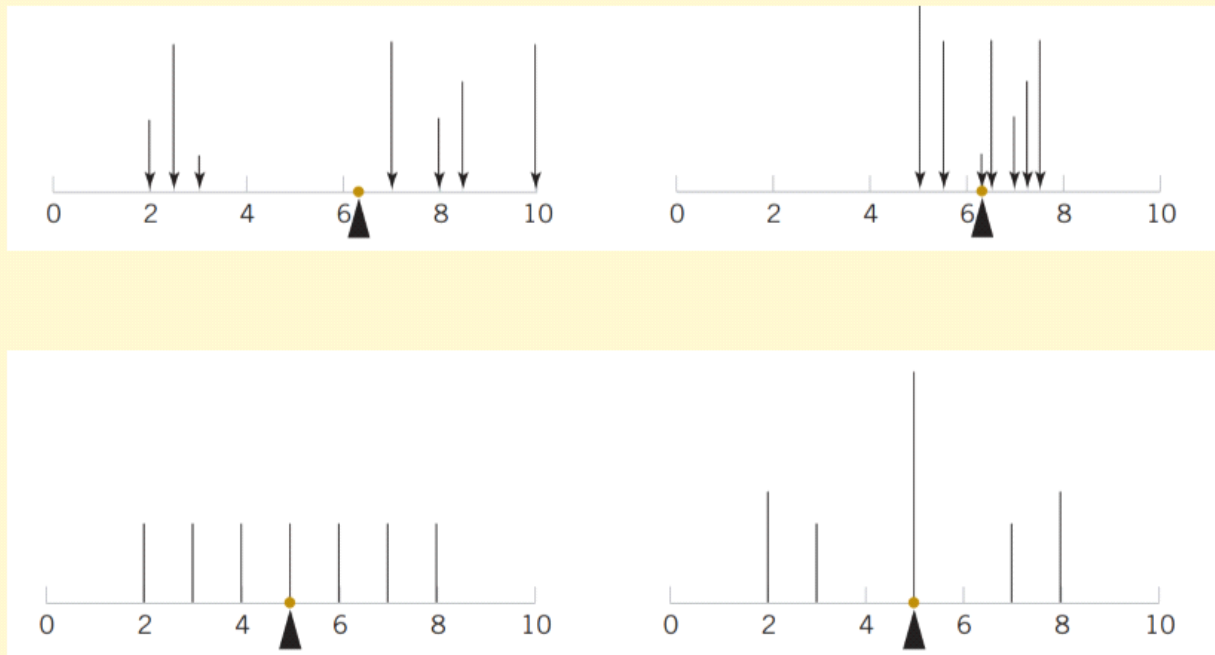
$$\frac{E[(X-\mu)^4]}{\sigma^4} - 3$$

↑
Benchmark

Excess Kurtosis > 0 , thick/heavy tail
 < 0 , light tail
 $= 0$, Normal



$\{\text{Mean, Variance}\} \neq \text{PMF/PDF}$



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Tyrannosaurus Rex (T-Rex)

Rex = King



Tyrannosaurus Rex (T-Rex)

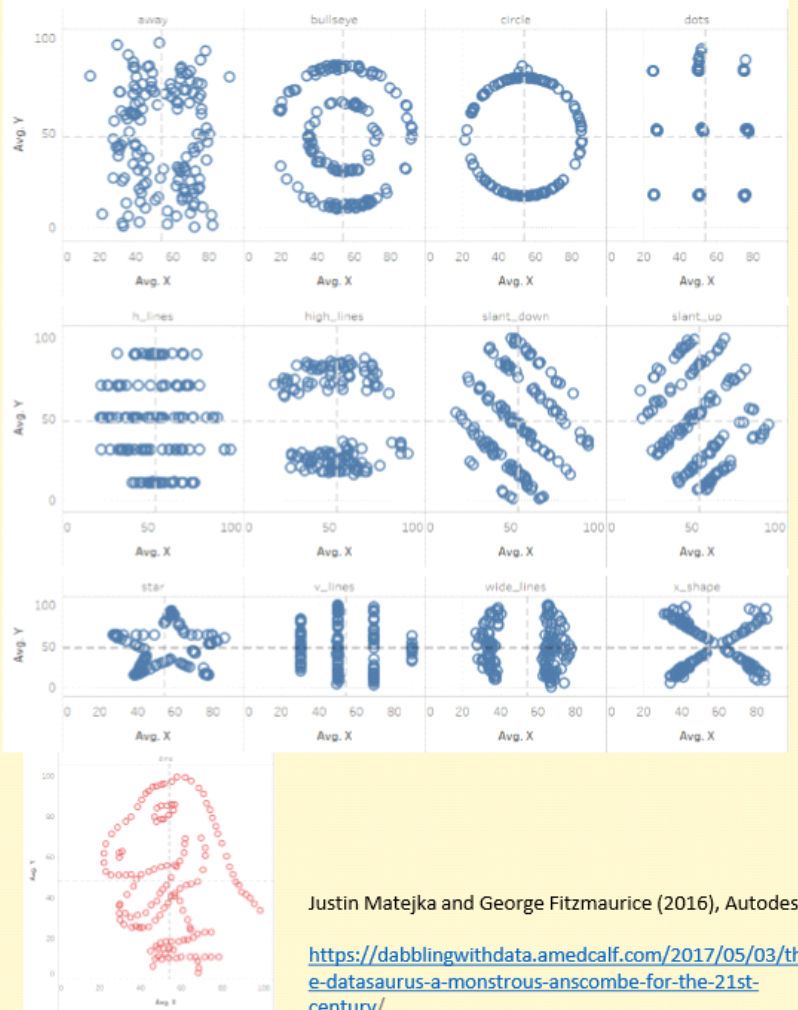


Thick tail

Datasaurus Dozen

Baker's dozen

Datasaurus Dozen



Justin Matejka and George Fitzmaurice (2016), Autodesk

<https://dabblingwithdata.amedcalf.com/2017/05/03/the-datasaurus-a-monstrous-anscombe-for-the-21st-century/>

Datasaurus Dozen

<https://seanfobbe.com/tutorials/importance-of-visualization-datasaurus/>

	dataset	x_mean	x_sd	y_mean	y_sd	corr
	<char>	<num>	<num>	<num>	<num>	<num>
1:	dino	54.26327	16.76514	47.83225	26.93540	-0.06447185
2:	away	54.26610	16.76982	47.83472	26.93974	-0.06412835
3:	h_lines	54.26144	16.76590	47.83025	26.93988	-0.06171484
4:	v_lines	54.26993	16.76996	47.83699	26.93768	-0.06944557
5:	x_shape	54.26015	16.76996	47.83972	26.93000	-0.06558334
6:	star	54.26734	16.76896	47.83955	26.93027	-0.06296110
7:	high_lines	54.26881	16.76670	47.83545	26.94000	-0.06850422
8:	dots	54.26030	16.76774	47.83983	26.93019	-0.06034144
9:	circle	54.26732	16.76001	47.83772	26.93004	-0.06834336
10:	bullseye	54.26873	16.76924	47.83082	26.93573	-0.06858639
11:	slant_up	54.26588	16.76885	47.83150	26.93861	-0.06860921
12:	slant_down	54.26785	16.76676	47.83590	26.93610	-0.06897974
13:	wide_lines	54.26692	16.77000	47.83160	26.93790	-0.06657523

$$\{\mu, \sigma^2\} \neq \text{PMF/PDF}$$

Datasaurus Dozen

